A Constructive Introduction to Quantum Field Theory

Lecture 2 — The Gaussian free field

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Notes: §3

Where we are

Last time. Positivity of the energy $\Rightarrow$ imaginary time; vacuum expectations are moments of $\mu(\mathrm{d}\varphi)$ " $= "$ $Z^{-1}\mathrm{e}^{-\int[\frac{1}{2}|\nabla\varphi|^2+\frac{1}{2}m^2\varphi^2+V(\varphi)]}\mathcal{D}\varphi$; relativistic QFTs $\leftrightarrow$ measures on $\mathcal{S}'(\mathbb{R}^d)$ satisfying the OS axioms.

Today. The Gaussian part, rigorously.

1. Construction on $\mathbb{T}^d$ and $\mathbb{R}^d$; regularity.
2. Reflection positivity; the free field; Källén–Lehmann again.
3. Local operators: Wick powers.
4. Gaussian integration by parts: the Dyson–Schwinger (DS) equation.
5. DS characterises the GFF.
6. The GFF is the stationary marginal of parabolic and elliptic stochastic quantisation.

Definition and construction

Setting and definition

$\mathcal{M} = \mathbb{T}_L^d = (\mathbb{R}/L\mathbb{Z})^d$ or $\mathbb{R}^d$; $\mathcal{S}(\mathcal{M})$ smooth, resp. Schwartz, functions; $\mathcal{S}'(\mathcal{M})$ with the cylinder (= Borel) σ-algebra.

$$A := m^2 - \Delta, \qquad C := A^{-1}, \qquad m > 0.$$

A, C are continuous bijections of $\mathcal{S}(\mathcal{M})$ (Fourier multipliers $(|k|^2 + m^2)^{\pm 1}$), and $\|C\|_{L^2 \to L^2} \leq m^{-2}$.

Definition

The Gaussian free field μ_m is the centred Gaussian measure on $\mathcal{S}'(\mathcal{M})$ with

$$\int \varphi(f)\varphi(g)\, \mu_m(\mathrm{d}\varphi) = \langle f, Cg \rangle, \qquad \text{i.e.} \qquad \int \mathrm{e}^{\mathrm{i}\varphi(f)} \mu_m(\mathrm{d}\varphi) = \mathrm{e}^{-\frac{1}{2}\langle f, Cf \rangle}.$$

Formally $\mu_m(\mathrm{d}\varphi)$ " = " $Z^{-1}\mathrm{e}^{-\frac{1}{2}\langle \varphi, A\varphi \rangle} \mathcal{D}\varphi$: the Gaussian part of the Euclidean action.

The Green function

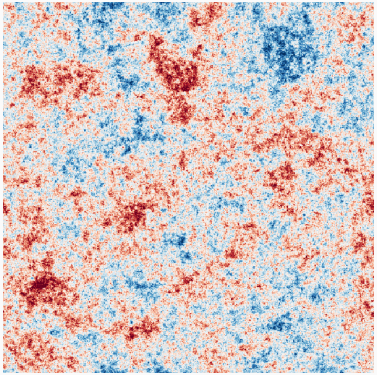
C has kernel G:

$$G_m(x) = \int \frac{e^{ik\cdot x}}{|k|^2 + m^2} \frac{dk}{(2\pi)^d}$$

on $\mathbb{R}^d$, and $\sum_{n\in\mathbb{Z}^d} G_m(x+nL)$ on $\mathbb{T}_L^d$.

- $d = 1$: $G_m(x) = e^{-m|x|}/(2m)$.
- $d = 2$: $G_m(x) \sim \frac{1}{2\pi} \log \frac{1}{|x|}$.
- $d \geq 3$: $G_m(x) \sim \frac{\Gamma(d/2-1)}{4\pi^{d/2}} |x|^{2-d}$.
- Decay $\approx e^{-m|x|}$ at infinity.

Singular on the diagonal for $d \geq 2$: the field is not a function.



A sample of the lattice GFF on $\mathbb{T}^2$ (512^2 sites).

Existence on the torus

Theorem

Let (e_n) be a real orthonormal basis of eigenfunctions of $-\Delta$ on $\mathbb{T}^d_L$, $-\Delta e_n = \kappa_n e_n$, $\lambda_n = m^2 + \kappa_n$, and (g_n) i.i.d. $N(0,1)$. For $s > \frac{d}{2} - 1$,

$$\varphi = \sum_n \lambda_n^{-1/2} g_n \, e_n$$

converges a.s. and in $L^2(\mathbb{P}; H^{-s})$, and $\mathrm{Law}(\varphi) = \mu_m$.

Proof.

$$\mathbb{E}\|\varphi\|^2_{H^{-s}} = \sum_n \frac{1}{(1+\kappa_n)^s \lambda_n} \asymp \sum_{k \in \mathbb{Z}^d} (1+|k|^2)^{-s-1} < \infty \iff 2s+2 > d.$$

L^2-bounded H^{-s}-valued martingale $\Rightarrow$ convergence. And $\varphi(f) = \sum_n \lambda_n^{-1/2} g_n \langle e_n, f\rangle$ is Gaussian with variance $\sum_n \lambda_n^{-1} \langle e_n, f\rangle^2 = \langle f, Cf\rangle$. □

Existence on $\mathbb{R}^d$ and regularity

Bochner–Minlos

$\Psi : \mathcal{S}(\mathbb{R}^d) \to \mathbb{C}$ continuous, positive definite, $\Psi(0) = 1 \Rightarrow$ there is a unique probability measure on $\mathcal{S}'(\mathbb{R}^d)$ with characteristic functional Ψ.

For $\Psi(f) = \mathrm{e}^{-\frac{1}{2}\langle f, Cf\rangle}$: continuity from $\langle f, Cf\rangle \le m^{-2}\|f\|_{L^2}^2$; positive definiteness from $\Psi(f_j - f_k) = a_j a_k \mathrm{e}^{\langle f_j, Cf_k\rangle}$ and the Schur product theorem.

Regularity (sharp, Exercise 3.3):

$d = 1$	$d = 2$	$d = 3$
stationary OU process	$H^{-\kappa}$, $\forall \kappa > 0$; not a function	$H^{-1/2-\kappa}$

The Cameron–Martin space H^1 (finite action) has measure zero.

Reflection positivity and the free field

Reflection positivity

$\mathcal{M} = \mathbb{R} \times N$, $N = \mathbb{R}^{d-1}$ or $\mathbb{T}_L^{d-1}$; $\Theta(\tau, \mathbf{x}) = (-\tau, \mathbf{x})$; $A = -\partial_\tau^2 + B$, $B = m^2 - \Delta_N$, $\omega = B^{1/2}$.

Proposition

(a) $C(\tau, \tau') = \mathrm{e}^{-|\tau - \tau'|\omega}(2\omega)^{-1}$ *(operator-valued kernel).*
(b) If $\operatorname{supp} f \subset \{\tau > 0\}$*:*

$$\langle \Theta f, Cf \rangle = \left\| (2\omega)^{-1/2} \int_0^\infty \mathrm{e}^{-\tau\omega} f(\tau, \cdot)\, \mathrm{d}\tau \right\|_{L^2(N)}^2 \geq 0.$$

(c) μ_m *on* $\mathbb{R}^d$ *satisfies OS0–OS4.*

Proof. (a) $\int \mathrm{e}^{\mathrm{i}kt}(k^2 + w^2)^{-1} \frac{\mathrm{d}k}{2\pi} = \frac{\mathrm{e}^{-w|t|}}{2w}$ and spectral calculus. (b) For $\tau < 0 < \tau'$, $|\tau - \tau'| = |\tau| + \tau'$: the kernel factorises. (c) OS3: $\hat{\mu}(f_i - \Theta f_j) = a_i a_j \mathrm{e}^{\langle \Theta f_i, C f_j \rangle}$; Schur. □

The free field, reconstructed

- OS reconstruction of μ_m gives the free field of mass m: Fock space over $L^2(N)$, $H = \mathrm{d}\Gamma(\omega)$ (Nelson's free Markov field).
- The time-zero field $\varphi(0,\cdot)$ exists and is Gaussian with covariance $(2\omega)^{-1}$: the ground state measure $\Omega_0^2 \mathcal{D}\varphi$ of the free Hamiltonian in the Schrödinger representation.
- The GFF is the continuum limit of the lattice spin system of Lecture 1 with $V = 0$ (Exercise 3.14).
- Markov property (Nelson): rooted in the locality of A.

Euclidean Källén–Lehmann

Wick rotation of $W_2^{(M)}$: $\int \mathrm{d}\Omega_M(\mathbf{p})\, \mathrm{e}^{-\omega_M(\mathbf{p})|\tau| + \mathrm{i}\mathbf{p}\cdot\mathbf{x}} = G_M(\tau, \mathbf{x})$. Hence for any theory satisfying the axioms

$$S_2(x) = v^2 + \int_{[0,\infty)} \rho(\mathrm{d}M^2)\, G_M(x), \qquad \hat{S}_2^T(k) = \int \frac{\rho(\mathrm{d}M^2)}{|k|^2 + M^2}.$$

The GFF is $\rho = \delta_{m^2}$.

- Short distances. G_M decreases in M and $G_M / G_m \to 1$ at 0 ($d \geq 2$), so $\liminf_{x\to 0} S_2^T(x)/G_m(x) \geq \rho([0,\infty)) > 0$: never less singular than free. Scale invariance $\Rightarrow$ unitarity bound $\Delta \geq \frac{d-2}{2}$ (Exercise 3.2).
- Long distances. $\operatorname{supp}\rho \subset [m_0^2, \infty) \Leftrightarrow$ decay at least like $\mathrm{e}^{-m_0|x|}$: mass gap = clustering.

Two masses: a positive but nonlocal theory

$\rho = c_1^2\delta_{m_1^2} + c_2^2\delta_{m_2^2}$, $0 < m_1 < m_2$: $C_\rho = c_1^2(m_1^2 - \Delta)^{-1} + c_2^2(m_2^2 - \Delta)^{-1}$, the law of $c_1\varphi_1 + c_2\varphi_2$ (independent GFFs). RP for each term $\Rightarrow$ OS axioms.

It reconstructs to two free species

$\mathcal{H} = \Gamma(\mathfrak{h}_{m_1}) \otimes \Gamma(\mathfrak{h}_{m_2})$, $\varphi = c_1\varphi_{m_1} \otimes \mathbb{1} + c_2\,\mathbb{1} \otimes \varphi_{m_2}$, $W_2 = c_1^2 W_2^{(m_1)} + c_2^2 W_2^{(m_2)}$.
Two kinds of free particles, created by the same field; no scattering (truncated $W_{n\geq 3} = 0$).

What is lost:

- $(\Box + m_1^2)(\Box + m_2^2)\varphi = 0$, but no Klein–Gordon equation; $(\varphi(0), \pi(0))$ are not complete Cauchy data.
- DS: $\mathbb{E}[D_h F] = \mathbb{E}[F\varphi(C_\rho^{-1}h)]$ with $C_\rho^{-1} = \dfrac{(m_1^2 - \Delta)(m_2^2 - \Delta)}{c_1^2(m_2^2 - \Delta) + c_2^2(m_1^2 - \Delta)}$ nonlocal; φ is not Markov.

Positivity versus locality

The local fourth-order action $\frac{1}{2}\langle\varphi,(m_1^2-\Delta)(m_2^2-\Delta)\varphi\rangle$ gives a Markov field with

$$\frac{1}{(|k|^2+m_1^2)(|k|^2+m_2^2)}=\frac{1}{m_2^2-m_1^2}\Big(\frac{1}{|k|^2+m_1^2}-\frac{1}{|k|^2+m_2^2}\Big),$$

so $\rho=(\delta_{m_1^2}-\delta_{m_2^2})/(m_2^2-m_1^2)$ has a negative weight.

- By Euclidean Källén–Lehmann it is not reflection positive: the mass m_2 is a ghost of negative norm (Pais–Uhlenbeck, Lee–Wick).
- Same for any $1/\hat{C}=$ polynomial of degree ≥ 2: $s\,\hat{C}(s)\to 0$, while $s\int\frac{\rho(\mathrm{d}M^2)}{s+M^2}\uparrow\rho([0,\infty))$.

Moral

For Gaussian fields, *positivity* (OS) and *locality* of the Euclidean field equation (C^{-1} a differential operator) together single out the free field: covariance $c(m^2-\Delta)^{-1}$ (or white noise).

Is the nonlocal theory physical?

- In Minkowski space, yes: all Wightman axioms hold, including locality (W3); positive energy, unique vacuum. Two free species: trivial and redundant, not inconsistent.
- "Nonlocal" is Euclidean: no Markov property $\Leftrightarrow$ φ alone is not a complete set of local degrees of freedom (no Cauchy problem, no Lagrangian in φ only). Cure: the pair (φ_1, φ_2) is local and Markov.
- Continuous ρ (a continuum of free species) is unphysical: no dynamical law of finite order, and too many localised states. Nuclearity (Haag–Swieca, Buchholz–Wichmann) needs $\sum_i n_i \mathrm{e}^{-\beta m_i} < \infty$: no sensible thermodynamics.
- Interacting theories also have a continuum in ρ, but produced by a *local* interaction: the Euclidean field equation of Φ^4_2, with $(m^2 - \Delta)\varphi + 4\lambda{:}\varphi^3{:}$ in the DS equation, is local.

Local operators: Wick products

Hermite polynomials

Interactions $\int V(\varphi(x))\mathrm{d}x$ need local functions of $\varphi(x)$: regularise, subtract, remove the regularisation.

$$\mathrm{e}^{ta-ct^2/2} = \sum_{n\geq 0} \frac{t^n}{n!} H_n(a;c) : \qquad H_2 = a^2 - c, \;\; H_3 = a^3 - 3ca, \;\; H_4 = a^4 - 6ca^2 + 3c^2.$$

Lemma

(X, Y) centred Gaussian, $\mathbb{E}X^2 = a$, $\mathbb{E}Y^2 = b$:

$$\mathbb{E}\big[H_n(X;a)H_k(Y;b)\big] = \delta_{nk}\, n!\, (\mathbb{E}[XY])^n.$$

Proof. $\mathbb{E}\big[\mathrm{e}^{sX - s^2a/2}\mathrm{e}^{tY - t^2b/2}\big] = \mathrm{e}^{st\,\mathbb{E}[XY]}$; compare coefficients of $s^n t^k$. □

Wick powers

Mollifier η_ε, $\varphi_\varepsilon = \eta_\varepsilon * \varphi$, $G_{\varepsilon,\delta} = \eta_\varepsilon * \eta_\delta * G$, $c_\varepsilon = G_{\varepsilon,\varepsilon}(0)$,

$$:\varphi_\varepsilon^n:(x) := H_n(\varphi_\varepsilon(x); c_\varepsilon), \qquad :\varphi_\varepsilon^n:(f) = \int f :\varphi_\varepsilon^n:.$$

Theorem

If $n(d-2) < d$ (all n in $d = 2$; $n \le 2$ in $d = 3$), then $:\varphi^n:(f) = \lim_{\varepsilon\to 0} :\varphi_\varepsilon^n:(f)$ exists in $L^2(\mu_m)$, is independent of η, and

$$\mathbb{E}\big[:\varphi^n:(f):\varphi^k:(g)\big] = \delta_{nk}\, n! \iint f(x) G(x-y)^n g(y)\,\mathrm{d}x\,\mathrm{d}y.$$

If $n(d-2) \ge d$ and $0 \le f \not\equiv 0$: $\mathbb{E}[:\varphi_\varepsilon^n:(f)^2] \to \infty$.

Proof

1. Uniform bound. $K(z) = 1 + \log_+ \frac{1}{|z|}$ $(d = 2)$, $1 + |z|^{2-d}$ $(d \geq 3)$. With $r = \max(\varepsilon, \delta)$:

- $|z| \geq 4r$: $G_{\varepsilon,\delta}(z) \leq \sup_{|w| \geq |z|/2} G(w) \leq CK(z)$;
- $|z| < 4r$: $G_{\varepsilon,\delta}(z) \leq \|\eta\|_\infty r^{-d} \int_{B(0,6r)} G \leq C \fint_{B(0,6r)} K \leq CK(r) \leq C'K(z)$.

2. Convergence. By the lemma,

$$\mathbb{E}\big[(:\varphi^n_\varepsilon:(f) - :\varphi^n_\delta:(f))^2\big] = n! \iint f(x)f(y)\big(G^n_{\varepsilon,\varepsilon} - 2G^n_{\varepsilon,\delta} + G^n_{\delta,\delta}\big)(x-y) \to 0$$

by dominated convergence: $K^n \in L^1_{\mathrm{loc}} \iff n(d-2) < d$.

3. Divergence. Otherwise $G^n \gtrsim |z|^{-n(d-2)} \notin L^1_{\mathrm{loc}}$; Fatou. □

Remarks

- Renormalisation: $\varphi_\varepsilon(x)^n$ has no limit; subtracting $c_\varepsilon \to \infty$ (via H_n) is the simplest counterterm. A different mollifier changes $:\varphi^n:$ by lower Wick powers with *finite* coefficients (Exercise 3.4).
- Chaos: $:\varphi^n:(f)$ lies in the n-th Wiener chaos; $L^2(\mu_m) \simeq$ Fock space over $H^{-1}(\mathcal{M})$. Hypercontractivity $\|X\|_{L^p} \le (p-1)^{n/2}\|X\|_{L^2}$ gives convergence in all L^p (and in $\mathcal{C}^{-\kappa}$ for $d = 2$).
- Towards Φ^4_2: on $\mathbb{T}^2$, $\int :\varphi^4: \in \bigcap_p L^p$ and (Nelson) $\mathrm{e}^{-\lambda \int :\varphi^4:} \in L^1(\mu_m)$.
- Other local operators: $\partial^\alpha \varphi$; in $d = 2$ the Wick exponentials $:\mathrm{e}^{\mathrm{i}\beta\varphi}:$, in L^2 iff $\beta^2 < 4\pi$ (sine-Gordon; Exercise 3.6).

The Dyson–Schwinger equation

Gaussian integration by parts

Cylinder functions $\mathcal{F}C_b^\infty \ni F(\varphi) = g(\varphi(f_1), \dots, \varphi(f_n))$, $g \in C_b^\infty$:

$$D_h F(\varphi) = \frac{\mathrm{d}}{\mathrm{d}t} F(\varphi + th)\Big|_{t=0} = \sum_i \partial_i g(\dots)\langle f_i, h\rangle = \langle DF, h\rangle, \qquad DF(x) = \frac{\delta F}{\delta \varphi(x)}.$$

Lemma

$(Y, X_1, \dots, X_n)$ *centred Gaussian:* $\mathbb{E}[Y\, g(X)] = \sum_i \mathbb{E}[YX_i]\, \mathbb{E}[\partial_i g(X)]$.

Proof. For $g = \mathrm{e}^{\mathrm{i}\langle u, \cdot\rangle}$: $\mathbb{E}[Y\mathrm{e}^{\mathrm{i}\langle u, X\rangle}] = -\mathrm{i}\partial_\theta \mathbb{E}[\mathrm{e}^{\mathrm{i}\theta Y + \mathrm{i}\langle u, X\rangle}]|_{\theta=0} = \mathrm{i}\sum_i u_i \mathbb{E}[YX_i]\mathbb{E}[\mathrm{e}^{\mathrm{i}\langle u, X\rangle}]$; then Fourier inversion and approximation. □

The Dyson–Schwinger equation

Proposition

For all $F \in \mathcal{FC}_b^\infty$, $h \in \mathcal{S}(\mathcal{M})$:

$$\mathbb{E}_{\mu_m}[D_h F] = \mathbb{E}_{\mu_m}\big[F\,\varphi(Ah)\big], \qquad \textit{formally} \qquad \mathbb{E}\Big[\frac{\delta F}{\delta\varphi(x)}\Big] = \mathbb{E}\big[F\,(m^2-\Delta)\varphi(x)\big].$$

Proof. $\mathbb{E}[\varphi(Ah)\varphi(f_i)] = \langle Ah, Cf_i\rangle = \langle h, f_i\rangle$; apply the lemma. □

With $F = \varphi(f_2)\cdots\varphi(f_n)$ one gets the hierarchy

$$(m^2-\Delta_{x_1})S_n(x_1,\dots,x_n) = \sum_{j=2}^{n}\delta(x_1-x_j)\,S_{n-2}(\dots\widehat{x_j}\dots),$$

which with $S_0 = 1$, $S_1 = 0$ gives Wick's theorem (Exercise 3.7).

DS = Euclidean field equation

- Minkowski: $(\Box + m^2)\varphi = 0$ and CCR. Euclidean: $(-\Delta + m^2)\varphi$ "=" $\delta/\delta\varphi$ inside expectations; the contact terms come from the CCR (time ordering).
- For a formal Gibbs measure $\mu_V \propto \mathrm{e}^{-\int V(\varphi)}\mu_m$:

$$\mathbb{E}_{\mu_V}[D_h F] = \mathbb{E}_{\mu_V}\Big[F\big((m^2 - \Delta)\varphi + V'(\varphi)\big)(h)\Big],$$

  the Euclidean version of $(\Box + m^2)\varphi + V'(\varphi) = 0$.
- $V'(\varphi)$ must be a well-defined local operator: Φ^4_2: $V' = 4\lambda{:}\varphi^3{:}$. For Φ^4_3 the renormalised DS equation is a theorem (Gubinelli–Hofmanová 2021).

DS characterises the GFF

Theorem

Let ν be a probability measure on $\mathcal{S}'(\mathcal{M})$ with $\varphi(f) \in L^1(\nu)$ for all f and

$$\mathbb{E}_\nu[D_h F] = \mathbb{E}_\nu[F\,\varphi(Ah)] \qquad \forall F \in \mathcal{F}C_b^\infty,\ h \in \mathcal{S}(\mathcal{M}).$$

Then $\nu = \mu_m$.

Proof. Fix f, take $h = Cf \in \mathcal{S}(\mathcal{M})$ and $F_s = \mathrm{e}^{\mathrm{i}s\varphi(f)}$, so $D_h F_s = \mathrm{i}s\langle f, Cf\rangle F_s$. With $\chi(s) = \mathbb{E}_\nu F_s$:

$$\mathrm{i}s\langle f, Cf\rangle \chi(s) = \mathbb{E}_\nu\big[\varphi(f)\mathrm{e}^{\mathrm{i}s\varphi(f)}\big] = -\mathrm{i}\chi'(s) \quad\Longrightarrow\quad \chi(s) = \mathrm{e}^{-\frac{s^2}{2}\langle f, Cf\rangle}.$$

So $\hat{\nu}(f) = \mathrm{e}^{-\frac{1}{2}\langle f, Cf\rangle}$ for all f; linearity of $f \mapsto \varphi(f)$ and Cramér–Wold give all finite-dimensional laws. □

Remarks

- The integrability assumption is exactly what is needed for DS to make sense: $\varphi(Ah)$ ranges over all $\varphi(g)$.
- DS for characteristic functions is a first-order linear ODE.
- $m = 0$ on the torus: *no* probability measure solves DS (take $h = 1$); work modulo constants (Exercise 3.9).
- Finite dimensions: DS says $\nabla(\mathrm{e}^S \nu) = 0$, so uniqueness is easy for any S (Exercise 3.8).
- Interacting theories: DS is a singular nonlinear relation; uniqueness is hard and largely open. Results for the exponential model in $d = 2$: De Vecchi–Gubinelli–Turra.

Parabolic stochastic quantisation

Langevin dynamics (Parisi–Wu 1981)

Finite dimensions: $\mathrm{d}X_t = -\nabla S(X_t)\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$ has generator $\mathcal{L} = \Delta - \nabla S \cdot \nabla$ and

$$\mathcal{L}^* \mathrm{e}^{-S} = \nabla \cdot \left(\nabla \mathrm{e}^{-S} + \mathrm{e}^{-S} \nabla S\right) = 0 :$$

the Gibbs measure $\mathrm{e}^{-S}\mathrm{d}x/Z$ is invariant.

Apply this to the Euclidean action, with a fictitious stochastic time t:

$$\partial_t \Phi = -(m^2 - \Delta)\Phi - V'(\Phi) + \sqrt{2}\,\xi \qquad \text{on } \mathbb{R}_+ \times \mathcal{M},$$

ξ space-time white noise: $\mathbb{E}[\xi(u)\xi(v)] = \langle u, v \rangle_{L^2(\mathbb{R}\times\mathcal{M})}$.

For $V = 0$: the stochastic heat equation, an infinite-dimensional Ornstein–Uhlenbeck process.

DS ⇒ invariance

Itô's formula: for $F = g(\varphi(f_1), \dots, \varphi(f_n))$ the generator is

$$\mathcal{L}F(\varphi) = \sum_{i,j} \langle f_i, f_j \rangle \, \partial_{ij} g(\dots) - \sum_i \varphi(Af_i) \, \partial_i g(\dots).$$

Proposition

$\mathbb{E}_{\mu_m}[G \, \mathcal{L}F] = -\mathbb{E}_{\mu_m}\big[\langle DG, DF \rangle_{L^2(\mathcal{M})}\big]$ *for* $F, G \in \mathcal{FC}_b^\infty$; *in particular* $\mathbb{E}_{\mu_m}[\mathcal{L}F] = 0$.

Proof. Apply DS to $k \, \partial_i g$ with $h = f_i$ and sum over i:

$$\sum_i \mathbb{E}[k \, \partial_i g \, \varphi(Af_i)] = \sum_{i,j} \langle f_i, f_j \rangle \mathbb{E}\big[\partial_j k \, \partial_i g + k \, \partial_{ij} g\big]. \qquad \square$$

Static (DS) and dynamic (Langevin) characterisations are two faces of the same integration by parts.

The stationary solution

Theorem

For $t \in \mathbb{R}$, $f \in \mathcal{S}(\mathcal{M})$ let $\Phi_t(f) = \sqrt{2}\,\xi(K_{t,f})$, $K_{t,f}(r,x) = \mathbb{1}_{r<t}\big(\mathrm{e}^{-(t-r)A}f\big)(x)$.

(a) $\mathbb{E}[\Phi_t(f)\Phi_s(g)] = \langle f, \mathrm{e}^{-|t-s|A}Cg\rangle$*: stationary, and* $\mathrm{Law}(\Phi_t) = \mu_m$ *for every* t.

(b) Φ *is a mild and weak solution: for* $s < t$, $\Phi_t(h) = \Phi_s(h) - \int_s^t \Phi_r(Ah)\mathrm{d}r + \sqrt{2}\,\xi(\mathbb{1}_{(s,t)} \otimes h)$.

(c) *From any initial datum* $\Psi_0 \in \mathcal{S}'(\mathcal{M})$ *independent of* ξ: $\mathbb{E}\,\mathrm{e}^{\mathrm{i}\Psi_t(f)} \to \mathrm{e}^{-\frac{1}{2}\langle f, Cf\rangle}$. *Hence* μ_m *is the unique invariant measure.*

Proof of (a). $\langle K_{t,f}, K_{s,g}\rangle = \int_{-\infty}^{t\wedge s}\langle f, \mathrm{e}^{-(t+s-2r)A}g\rangle\mathrm{d}r = \langle f, \mathrm{e}^{-|t-s|A}(2A)^{-1}g\rangle$; an $\mathcal{S}'$-valued version exists by Minlos regularisation.

Proof of (b), (c)

(a), on $\mathbb{T}^d$. The modes $\Phi_t(e_n)$ are independent stationary OU processes $\mathrm{d}X = -\lambda_n X \mathrm{d}t + \sqrt{2}\mathrm{d}B$ with variance λ_n^{-1}.

(b) Mild form from the semigroup identity $K_{t,h} = K_{s,\mathrm{e}^{-(t-s)A}h} + \mathbb{1}_{(s,t)}\mathrm{e}^{-(t-\cdot)A}h$; weak form by stochastic Fubini (Exercise 3.12).

(c) By independence,

$$\mathbb{E}\,\mathrm{e}^{\mathrm{i}\Psi_t(f)} = \mathbb{E}\left[\mathrm{e}^{\mathrm{i}\Psi_0(\mathrm{e}^{-tA}f)}\right]\mathrm{e}^{-\frac{1}{2}\langle f,(1-\mathrm{e}^{-2tA})Cf\rangle}.$$

For $m > 0$, $\mathrm{e}^{-tA}f \to 0$ in $\mathcal{S}(\mathcal{M})$, so $\Psi_0(\mathrm{e}^{-tA}f) \to 0$ a.s. If $\Psi_0 \sim \mu_m$ the product is exactly $\mathrm{e}^{-\frac{1}{2}\langle f,Cf\rangle}$ (invariance); an invariant ν must have the same characteristic functional. □

Comments

- As a field on $\mathbb{R} \times \mathcal{M}$, the stationary solution has covariance $2(\partial_t + A)^{-1}(-\partial_t + A)^{-1} = 2(-\partial_t^2 + A^2)^{-1}$; its equal-time slice is $2\int_{\mathbb{R}} (\theta^2 + A^2)^{-1} \frac{\mathrm{d}\theta}{2\pi} = A^{-1}$.
- The stochastic time t has nothing to do with the Euclidean time τ.
- Interacting case: a singular SPDE.
  - Φ^4_2: Da Prato–Debussche 2003 (also Jona-Lasinio–Mitter, Albeverio–Röckner).
  - Φ^4_3: Hairer 2014; Gubinelli–Imkeller–Perkowski 2015; Catellier–Chouk 2018.
  - Global bounds: Mourrat–Weber 2017; Gubinelli–Hofmanová 2019; construction of Euclidean Φ^4_3 via the dynamics: Gubinelli–Hofmanová 2021.

Elliptic stochastic quantisation

Dimensional reduction (Parisi–Sourlas 1979)

The elliptic SPDE in $d+2$ dimensions

$$(m^2 - \Delta_{\mathbb{R}^2 \times \mathcal{M}})\Psi + V'(\Psi) = \zeta, \qquad \zeta \text{ white noise on } \mathbb{R}^2 \times \mathcal{M},$$

is related to the d-dimensional Gibbs measure: formally

$$\mathrm{Law}\big(\Psi(z,\cdot)\big) \propto \exp\Big(-4\pi\Big[\tfrac{1}{2}\langle \varphi, A\varphi\rangle + \int V(\varphi)\Big]\Big)\mathcal{D}\varphi.$$

- Mechanism: a hidden supersymmetry (Parisi–Sourlas; Klein–Landau–Perez 1984).
- Rigorous nonlinear results: $d = 0$ (Albeverio–De Vecchi–Gubinelli 2020); polynomial on $\mathbb{T}^2$ and exponential on $\mathbb{R}^2$ (ADG 2021); elliptic Φ^4 in 4 and 5 dimensions (Gubinelli–Hofmanová 2019).
- For $V = 0$ it is elementary, and it explains the 4π.

The linear case

Theorem

Let $\mathsf{L} = m^2 - \Delta_{\mathbb{R}^2 \times \mathcal{M}} = -\Delta_{\mathbb{R}^2} + A$ *and* ζ *white noise on* $\mathbb{R}^2 \times \mathcal{M}$.

(a) $\mathsf{L}\Psi = \sqrt{4\pi}\,\zeta$ *has a unique tempered solution* $\Psi(u) = \sqrt{4\pi}\,\zeta(\mathsf{L}^{-1}u)$.

(b) $K_{z,f} = \mathsf{L}^{-1}(\delta_z \otimes f) \in L^2$, *continuous in* z; $\Psi(z,f) = \sqrt{4\pi}\,\zeta(K_{z,f})$ *is the section of* Ψ *at* z.

(c) $\mathbb{E}[\Psi(z,f)\Psi(z,g)] = \langle f, Cg\rangle$: *every section has law* μ_m, *and* $(\Psi(z,\cdot))_{z\in\mathbb{R}^2}$ *is stationary and isotropic.*

Uniqueness: L maps $\mathcal{S}$ onto $\mathcal{S}$, so $D(\mathsf{L}u) = 0\ \forall u$ forces $D = 0$.

Proof: where 4π comes from

Fourier transform in z only: $\widehat{K_{z_f}}(q) = \mathrm{e}^{-\mathrm{i}q\cdot z}(|q|^2 + A)^{-1} f$. By Plancherel and the spectral theorem,

$$\langle K_{z_f}, K_{z_g} \rangle = \Big\langle f, \Big[\int_{\mathbb{R}^2} \frac{\mathrm{d}q}{(2\pi)^2} (|q|^2 + A)^{-2} \Big] g \Big\rangle = \frac{1}{4\pi} \langle f, A^{-1} g \rangle,$$

since

$$\int_{\mathbb{R}^2} \frac{\mathrm{d}q}{(2\pi)^2} \frac{1}{(|q|^2 + a)^2} = \frac{1}{2\pi} \int_0^\infty \frac{r\,\mathrm{d}r}{(r^2 + a)^2} = \frac{1}{4\pi a}.$$

Multiply by 4π. Stationarity and isotropy: the covariance $4\pi \int \mathrm{e}^{\mathrm{i}q\cdot(z-z')} \langle f, (|q|^2 + A)^{-2} g \rangle \frac{\mathrm{d}q}{(2\pi)^2}$ depends only on $|z - z'|$. □

Slices of Gaussian fields

Field on $\mathbb{R}^k \times \mathcal{M}$ with covariance $(-\Delta_{\mathbb{R}^k} + A)^{-p}$; its slice $\{0\} \times \mathcal{M}$ has covariance

$$c_{k,p}\, A^{k/2-p}, \qquad c_{k,p} = \frac{\Gamma(p - \frac{k}{2})}{(4\pi)^{k/2}\Gamma(p)}.$$

(k,p)	slice covariance	interpretation
$(1,1)$	$\frac{1}{2}B^{-1/2} = (2\omega)^{-1}$	time-zero field / free vacuum
$(2,2)$	$(4\pi)^{-1}A^{-1}$	elliptic stochastic quantisation
parabolic	$2(-\partial_t^2 + A^2)^{-1} \;\to\; A^{-1}$	Langevin dynamics
$(1,2)$	$\propto A^{-3/2}$	not a GFF (e.g. codim-1 slices of the elliptic field)

The GFF arises iff $p = 1 + \frac{k}{2}$. For $\mathcal{M} = \mathbb{R}^d$, every codimension-two slice of the elliptic field is a GFF.

Outlook

Next lectures

- Lecture 3: the GFF along a scale parameter; stochastic observables; Polchinski equation and its FBSDE; Wetterich equation; variational structure.
- Lecture 4: Φ^4_2 — Wick-ordered interaction, uniform bounds, infinite volume, renormalised DS equation.
- Lecture 5: Φ^4_3 — counterterms and the renormalised variational problem. Lecture 6: perspectives.

The pattern

Gaussian measure $\xrightarrow{\text{local perturbation}}$ renormalised DS equation / scale dynamics $\xrightarrow{\text{estimates}}$ Euclidean QFT $\xrightarrow{\text{OS}}$ relativistic QFT.

Exercises and reading

Exercises (numbering of the notes)

- 3.2 Unitarity bound.
- 3.3 Regularity is sharp.
- 3.6 Wick exponentials, $\beta^2 < 4\pi$.
- 3.7 Wick's theorem from DS.
- 3.8 DS in finite dimensions.
- 3.10 Infinitesimal invariance characterises μ_m.
- 3.13 Slices of Gaussian fields.
- 3.15 Computer experiments.
- 3.16 Two masses: positivity vs. locality.

Reading

- Sheffield, *GFF for mathematicians*, PTRF 2007.
- Janson, *Gaussian Hilbert Spaces*.
- Glimm–Jaffe; Simon, $P(\varphi)_2$.
- Da Prato–Zabczyk, *Stochastic Equations in Infinite Dimensions*.
- Albeverio–De Vecchi–Gubinelli, *Elliptic stochastic quantization*, AoP 2020.